

Approximating the Average-Case Graph Search Problem with Non-Uniform Costs

Michał Szyfelbein

Faculty of Electronics, Telecommunications and Informatics
Gdańsk University of Technology, Poland

December 29, 2025

Binary Search

Binary Search – a classical strategy used to efficiently locate a hidden **target** element x in a linearly ordered set using $O(\log n)$ comparison operations.

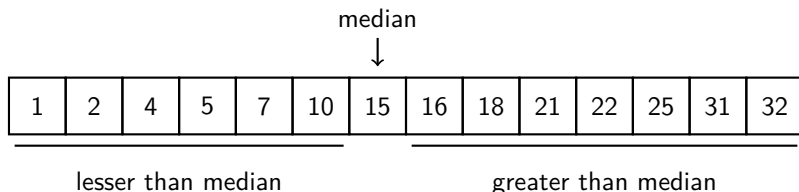


Figure: Example of a sorted array containing 14 elements.

Searching in Graphs

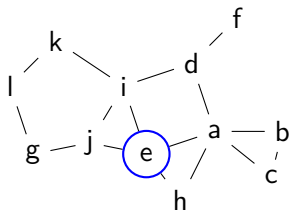
Easy to generalize to graphs:

A **query** to a vertex v returns information whether v is the target x , and if not, which connected component of $G - v$ contains x .

Searching in Graphs

Easy to generalize to graphs:

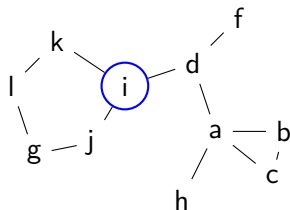
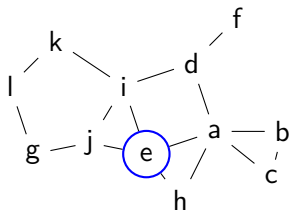
A **query** to a vertex v returns information whether v is the target x , and if not, which connected component of $G - v$ contains x .



Searching in Graphs

Easy to generalize to graphs:

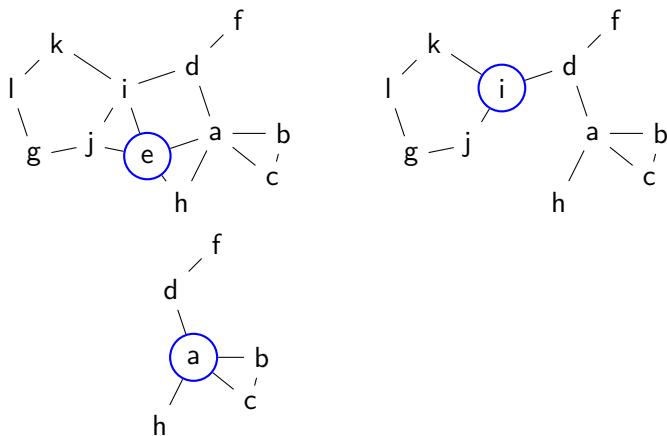
A **query** to a vertex v returns information whether v is the target x , and if not, which connected component of $G - v$ contains x .



Searching in Graphs

Easy to generalize to graphs:

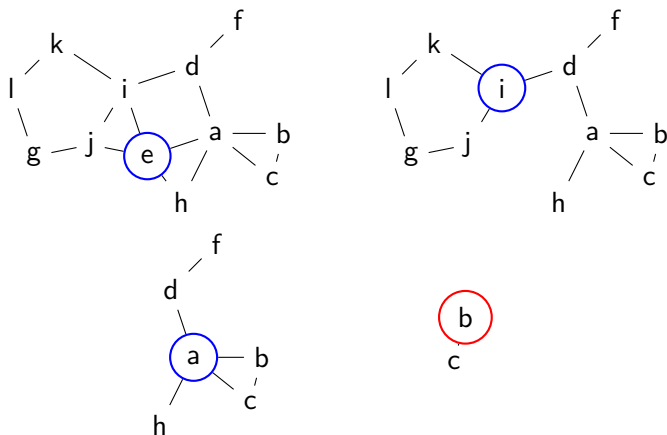
A **query** to a vertex v returns information whether v is the target x , and if not, which connected component of $G - v$ contains x .



Searching in Graphs

Easy to generalize to graphs:

A **query** to a vertex v returns information whether v is the target x , and if not, which connected component of $G - v$ contains x .



Strategy of searching

We wish to find a strategy of searching.

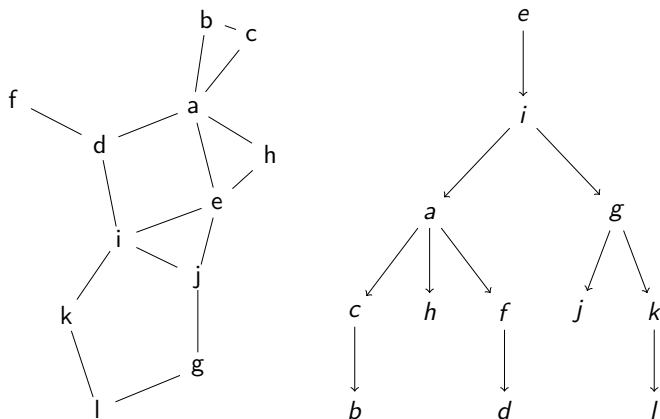


Figure: Sample input graph a decision tree for it.

Additional parameters

We introduce additional information about the input:

- ▶ To each vertex v we assign an arbitrary **cost** $c(v)$, which denotes a cost of performing a query to v .
- ▶ Additionally, to each vertex v we also assign an arbitrary **weight** $w(v)$, which denotes the importance of v .

Our setup

What is the best strategy of searching in a graph?

Graph Search Problem (GSP)

Input: Graph G , a query cost function $c: V(G) \rightarrow \mathbb{N}$ and a weight function $w: V(G) \rightarrow \mathbb{N}$.

Output: A decision tree D minimizing the weighted average search cost:

$$c_G(D) = \sum_{x \in V(G)} w(x) \cdot c(Q_G(D, x))$$

where $Q_G(D, x)$ denotes the set of queries performed along the unique path in D from the root $r(D)$ to x .

How to tackle the problem

Bad news: The Graph Search Problem is **NP-hard** even when restricted to bounded degree trees and bounded diameter trees. We want to find an algorithm providing a good **approximation** for the problem:

- ▶ $(4 + \epsilon)$ -approximation for **trees**.
- ▶ $O(\sqrt{\log n})$ -approximation for **general graphs**.

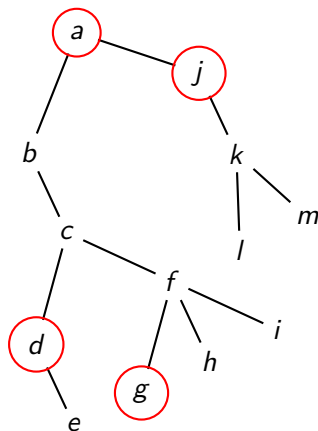
Weighted α -Separator Problem

Weighted α -Separator Problem

Input: Graph G , a cost function $c: V \rightarrow \mathbb{N}$, a weight function $w: V \rightarrow \mathbb{N}$ and a real number α .

Output: A set $S \subseteq V(G)$ called **separator** such that for every $H \in G - S$, $w(H) \leq w(G)/\alpha$ and $c(S)$ is minimized.

Example of a separator



	$w(v)$	$c(v)$
a	2	3
b	1	4
c	3	6
d	2	2
e	4	1
f	0	3
g	1	1
h	4	3
i	2	3
j	5	2
k	1	2
l	2	3
m	3	4

Figure: Sample input tree T and a weighted 3-separator (the circled vertices) of cost 8.

How to find the separator

Bad news again: The Weighted α -separator Problem is **NP-hard** as well. But there exists a bicriteria **FPTAS** for trees:

Theorem

Let S be an optimal weighted α -separator for (T, c, w, α) . For any $\delta > 0$ there exists an algorithm `SeparatorFPTAS`, which returns a separator S' , such that:

1. $c(S') \leq c(S)$.
2. $w(H) \leq \frac{(1+\delta) \cdot w(T)}{\alpha}$ for every $H \in T - S'$.
3. The algorithm runs in $O(n^3/\delta^2)$ time.

Notation

To connect searching and separating we need the following notation:

- ▶ $\mathcal{R}_D(G)$ – the family of all candidate subsets of D in G .
- ▶ D^* – optimal decision tree.
- ▶ $\mathcal{L}_k^*$ – the k -th **level** of $\text{OPT}(G)$: the subset of $\mathcal{R}_{D^*}(G)$ consisting of all maximal elements H of $\mathcal{R}_{D^*}(G)$ with $w(H) \leq k$.
- ▶ $S_k^* = V(G) - \mathcal{L}_k^*$ – vertices belonging to the separator at the level $\mathcal{L}_k^*$. S_k^* forms a weighted $w(G)/k$ -separator of G .

Example of the connection

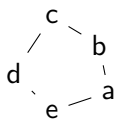


Figure: G .



Figure: D^* .

Example of the connection

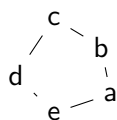


Figure: G .

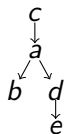


Figure: D^* .

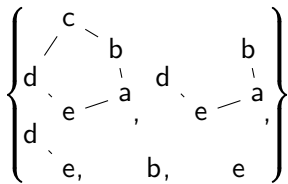


Figure: $\mathcal{R}_{D^*}(G)$.

Example of the connection

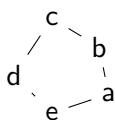


Figure: G .



Figure: D^* .

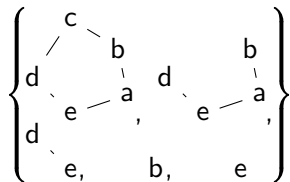


Figure: $\mathcal{R}_{D^*}(G)$.

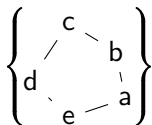


Figure: $\mathcal{L}_5^*, S_5^* = \{\}$.

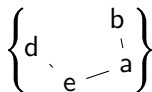


Figure: $\mathcal{L}_4^*, S_4^* = \{c\}$.

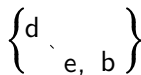


Figure: $\mathcal{L}_3^*, S_3^* = \{a, c\}$.

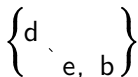


Figure: $\mathcal{L}_2^*, S_2^* = \{a, c\}$.



Figure:



Figure:

$\mathcal{L}_0^*, S_0^* = \{a, b, c, d, e\}$.

Basic lemmas

Lemma

Let $G_{D,v}$ be the candidate subgraph of G in which v is queried when using D . Then, $c_G(D) = \sum_{v \in V(G)} w(G_{D,v}) \cdot c(v)$.

Basic lemmas

Lemma

Let $G_{D,v}$ be the candidate subgraph of G in which v is queried when using D . Then, $c_G(D) = \sum_{v \in V(G)} w(G_{D,v}) \cdot c(v)$.

Lemma

$$\text{OPT}(G) = \sum_{k=0}^{w(G)-1} c(S_k^*).$$

Basic lemmas

Lemma

$$2 \cdot OPT(G) = 2 \cdot \sum_{k=0}^{w(G)-1} c(S_k^*) \geq \sum_{k=0}^{w(G)} c(S_{\lfloor k/2 \rfloor}^*).$$

Basic lemmas

Lemma

$$2 \cdot \text{OPT}(G) = 2 \cdot \sum_{k=0}^{w(G)-1} c(S_k^*) \geq \sum_{k=0}^{w(G)} c(S_{\lfloor k/2 \rfloor}^*).$$

Lemma

Let $\mathcal{G}$ be any subgraph of G and $0 \leq \beta \leq 1$. Then:

$$\beta \cdot w(\mathcal{G}) \cdot c(S_{\lfloor w(\mathcal{G})/2 \rfloor}^* \cap \mathcal{G}) \leq \sum_{k=(1-\beta)w(\mathcal{G})+1}^{w(\mathcal{G})} c(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{G}).$$

Searching in Trees

We will iteratively use the **FPTAS** for the Weighted α -separator problem to create an $(4 + \epsilon)$ -approximation algorithm for the Tree Search Problem:

Theorem

For any $\epsilon > 0$ there exists an $(4 + \epsilon)$ -approximation algorithm for the Tree Search Problem running in $O(n^4/\epsilon^2)$ time.

The algorithm

proc DecisionTree(T, c, w, ϵ):

1. $S_T \leftarrow \text{SeparatorFPTAS}\left(T, c, w, \alpha = 2, \delta = \frac{\epsilon}{4+\epsilon}\right)$.
2. $D_T \leftarrow$ arbitrary partial decision tree for T , built from vertices of S_T .
3. For each $H \in T - S_T$:
 - 3.1 $D_H \leftarrow \text{DecisionTree}(H, c, w, \epsilon)$.
 - 3.2 Hang D_H in D_T below the last query to $v \in N_T(H)$.
4. Return D_T .

Structure of the solution

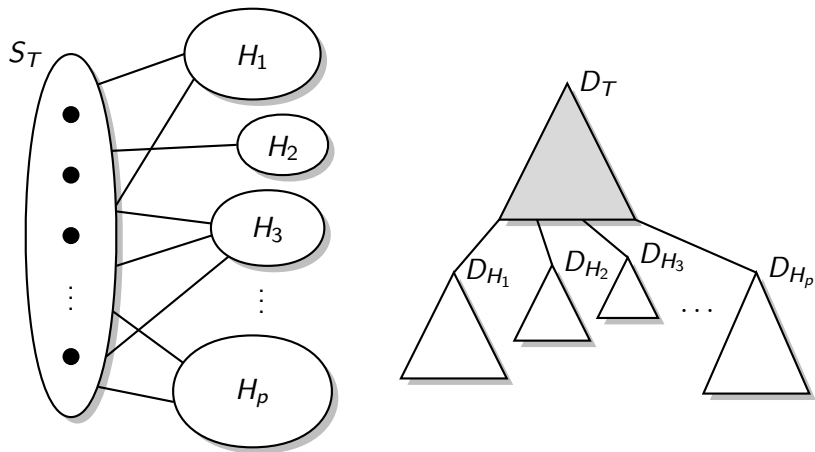


Figure: The separator S_T produced by the algorithm and the structure of the decision tree built using S_T .

Sketch of the proof

- ▶ $\mathcal{T}$ – subtree of T for which the procedure was called.
- ▶ Let $S_{\mathcal{T}}^* = S_{\lfloor w(\mathcal{T})/2 \rfloor}^* \cap \mathcal{T}$. We have $c(S_{\mathcal{T}}) \leq c(S_{\mathcal{T}}^*)$.
- ▶ We have that the contribution of the decision tree $D_{\mathcal{T}}$ is bounded by:

Sketch of the proof

- ▶ $\mathcal{T}$ – subtree of T for which the procedure was called.
- ▶ Let $S_{\mathcal{T}}^* = S_{\lfloor w(\mathcal{T})/2 \rfloor}^* \cap \mathcal{T}$. We have $c(S_{\mathcal{T}}) \leq c(S_{\mathcal{T}}^*)$.
- ▶ We have that the contribution of the decision tree $D_{\mathcal{T}}$ is bounded by:

$$\begin{aligned} w(\mathcal{T}) \cdot c(S_{\mathcal{T}}) &\leq w(\mathcal{T}) \cdot c(S_{\mathcal{T}}^*) \\ &\leq \frac{2}{1-\delta} \cdot \sum_{k=\frac{1+\delta}{2} \cdot w(\mathcal{T})+1}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{T}\right). \end{aligned}$$

Sketch of the proof

- ▶ Let $\beta = \frac{1-\delta}{2}$. We have:

$$\begin{aligned} c_T(D) &\leq \frac{2}{1-\delta} \cdot \sum_{\mathcal{T}} \sum_{k=\frac{1+\delta}{2} \cdot w(\mathcal{T})+1}^{w(\mathcal{T})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{T}\right) \\ &\leq \frac{2}{1-\delta} \cdot \sum_{k=0}^{w(T)} c\left(S_{\lfloor k/2 \rfloor}^*\right) \leq (4 + \epsilon) \cdot \text{OPT}(T). \end{aligned}$$

- ▶ As $1/\delta = \frac{4+\epsilon}{\epsilon} = 1 + 4/\epsilon$, the running time is $O(n^4/\epsilon^2)$.

Min-Ratio Vertex Cut Problem

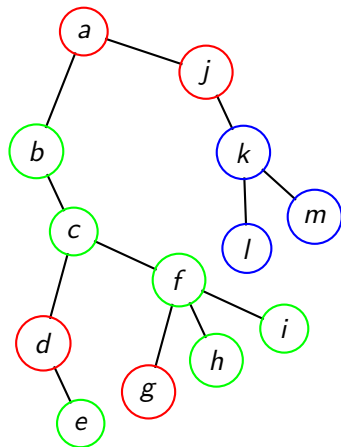
Min-Ratio Vertex Cut Problem

Input: Graph $G = (V(G), E(G))$, the cost function $c: V \rightarrow \mathbb{N}$ and the weight function $w: V \rightarrow \mathbb{N}$.

Output: A partition (A, S, B) of $V(G)$ called **vertex-cut**, such that there are no $u \in A$ and $v \in B$ for which $uv \in E(G)$, minimizing the ratio:

$$\alpha_{c,w}(A, S, B) = \frac{c(S)}{w(A \cup S) \cdot w(B \cup S)}.$$

Example of a vertex cut



	$w(v)$	$c(v)$
a	2	3
b	1	4
c	3	6
d	2	2
e	4	1
f	0	3
g	1	1
h	4	3
i	2	3
j	5	2
k	1	2
l	2	3
m	3	4

Figure: A tree and a vertex cut of ratio

$$\alpha_{c,w}(A, S, B) = \frac{c(S)}{w(A \cup S) \cdot w(B \cup S)} = \frac{8}{(6+10) \cdot (14+10)} = \frac{8}{384} = \frac{1}{48}.$$

How to find the cut

Min-Ratio Vertex Cut Problem is also **NP-hard**. However, we use the following result of [FHL05]:

Theorem

Given a graph $G = (V(G), E(G))$, the cost function $c : V \rightarrow \mathbb{N}$ and the weight function $w : V \rightarrow \mathbb{N}$, there exists a polynomial-time algorithm, which computes a partition (A, S, B) , such that:

$$\alpha_{c,w}(A, S, B) = O\left(\sqrt{\log n}\right) \cdot \alpha_{c,w}(G).$$

Searching in Graphs

We will iteratively use the f_n -approximation algorithm for the Min-Ratio Vertex Cut Problem to create an $O(f_n)$ -approximation algorithm for the Graph Search Problem:

Theorem

Let f_n be the approximation ratio of any polynomial time algorithm for the Min-Ratio Vertex Cut Problem. Then, there exists an $O(f_n)$ -approximation algorithm for the Graph Search Problem, running in polynomial time.

The algorithm

proc DecisionTree(T, c, w, ϵ):

1. $A_G, S_G, B_G \leftarrow \text{AlgorithmMinCut}(G, c, w)$.
2. $D_G \leftarrow$ arbitrary partial decision tree for G , built from vertices of S_G .
3. For each $H \in G - S_G$:
 - 3.1 $D_H \leftarrow \text{DecisionTree}(H, c, w)$.
 - 3.2 Hang D_H in D_G below the last query to $v \in N_G(H)$.
4. Return D_G .

Sketch of the proof

- ▶ $\mathcal{G}$ – any subgraph of G , for which the procedure was called.
- ▶ Let $S_{\mathcal{G}}^* = S_{\lfloor w(\mathcal{G})/2 \rfloor}^* \cap \mathcal{G}$.

Sketch of the proof

- ▶ $\mathcal{G}$ – any subgraph of G , for which the procedure was called.
- ▶ Let $S_{\mathcal{G}}^* = S_{\lfloor w(\mathcal{G})/2 \rfloor}^* \cap \mathcal{G}$.

Lemma

Let $\mathcal{H} = \mathcal{G} - S_{\mathcal{G}}^*$. Then, we can partition $\mathcal{H}$ into two sets, $\mathcal{A}$ and $\mathcal{B}$ such that for $A = \bigcup_{H \in \mathcal{A}} V(H)$ and $B = \bigcup_{H \in \mathcal{B}} V(H)$, we have:

$$w(A \cup S_{\mathcal{G}}^*) \cdot w(B \cup S_{\mathcal{G}}^*) \geq w(\mathcal{G})^2 / 11.$$

The partition $(A, S_{\mathcal{G}}^*, B)$ in the above lemma is a vertex cut of $\mathcal{G}$, so we have:

$$\alpha_{c,w}(\mathcal{G}) \leq \frac{c(S_{\mathcal{G}}^*)}{w(A \cup S_{\mathcal{G}}^*) \cdot w(B \cup S_{\mathcal{G}}^*)} \leq \frac{11 \cdot c(S_{\mathcal{G}}^*)}{w(\mathcal{G})^2}.$$

Sketch of the proof

- ▶ Let $(A_{\mathcal{G}}, S_{\mathcal{G}}, B_{\mathcal{G}})$, be the partition returned by the algorithm:

$$\alpha_{c,w}(A_{\mathcal{G}}, S_{\mathcal{G}}, B_{\mathcal{G}}) = \frac{c(S_{\mathcal{G}})}{w(A_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}})} \leq f_n \cdot \frac{11 \cdot c(S_{\mathcal{G}}^*)}{w(\mathcal{G})^2}.$$

- ▶ Let $\beta = w(B_{\mathcal{G}} \cup S_{\mathcal{G}}) / w(\mathcal{G})$, so that $(1 - \beta) \cdot w(\mathcal{G}) = w(A_{\mathcal{G}})$.
- ▶ Assume w. l. o. s. that $w(A_{\mathcal{G}}) \geq w(B_{\mathcal{G}})$.

Sketch of the proof

- ▶ Let $(A_{\mathcal{G}}, S_{\mathcal{G}}, B_{\mathcal{G}})$, be the partition returned by the algorithm:

$$\alpha_{c,w}(A_{\mathcal{G}}, S_{\mathcal{G}}, B_{\mathcal{G}}) = \frac{c(S_{\mathcal{G}})}{w(A_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}})} \leq f_n \cdot \frac{11 \cdot c(S_{\mathcal{G}}^*)}{w(\mathcal{G})^2}.$$

- ▶ Let $\beta = w(B_{\mathcal{G}} \cup S_{\mathcal{G}}) / w(\mathcal{G})$, so that $(1 - \beta) \cdot w(\mathcal{G}) = w(A_{\mathcal{G}})$.
- ▶ Assume w. l. o. s. that $w(A_{\mathcal{G}}) \geq w(B_{\mathcal{G}})$.
- ▶ The contribution of the decision tree $D_{\mathcal{G}}$ is bounded by:

$$\begin{aligned} w(\mathcal{G}) \cdot c(S_{\mathcal{G}}) &\leq 11 \cdot f_n \cdot \frac{w(A_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}})}{w(\mathcal{G})} \cdot c(S_{\mathcal{G}}^*) \\ &\leq 11 \cdot f_n \cdot w(B_{\mathcal{G}} \cup S_{\mathcal{G}}) \cdot c(S_{\mathcal{G}}^*) \\ &\leq 11 \cdot f_n \cdot \sum_{k=w(A_{\mathcal{G}})+1}^{w(\mathcal{G})} c(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{G}) \end{aligned}$$

Sketch of the proof

Let D be the decision tree returned by the procedure. We have:

$$\begin{aligned}c_G(D) &\leq \sum_{\mathcal{G}} w(\mathcal{G}) \cdot c(S_{\mathcal{G}}) \\&\leq 11 \cdot f_n \cdot \sum_{\mathcal{G}} \sum_{k=w(A_{\mathcal{G}})+1}^{w(\mathcal{G})} c\left(S_{\lfloor k/2 \rfloor}^* \cap \mathcal{G}\right) \\&\leq 11 \cdot f_n \cdot \sum_{k=0}^{w(G)} c\left(S_{\lfloor k/2 \rfloor}^*\right) \\&\leq 22 \cdot f_n \cdot \text{OPT}(G)\end{aligned}$$

Thank you for your attention!

Questions?

Bibliography I

- [FHL05] Uriel Feige, MohammadTaghi Hajiaghayi, and James R. Lee. “Improved approximation algorithms for minimum-weight vertex separators”. In: *Proceedings of the Thirty-Seventh Annual ACM Symposium on Theory of Computing*. STOC '05. Baltimore, MD, USA: Association for Computing Machinery, 2005, pp. 563–572. ISBN: 1581139608. DOI: 10.1145/1060590.1060674. URL: <https://doi.org/10.1145/1060590.1060674>.